NHS

MODEL ED OPTIMISER

Run tomorrow's emergency department today

A digital twin of **your** ED - flow, workforce and money in one place - so every change is tested in the model before it is tested on patients.

Built on the NHS England Model Emergency Department · runs on locked-down NHS networks · every assumption visible and editable

THE PROBLEM

The questions every ED leadership team is asked

- "Will we hold the **4-hour position** this winter, and what would it actually take?"
- "Why are ambulances **queuing at the door**, and which change frees them fastest?"
- "Is the **rota** matched to the demand curve, or are we paying agency at the wrong hours?"
- "Which investment - SDEC, UTC, frailty unit, beds - is **worth it**, in pounds?"

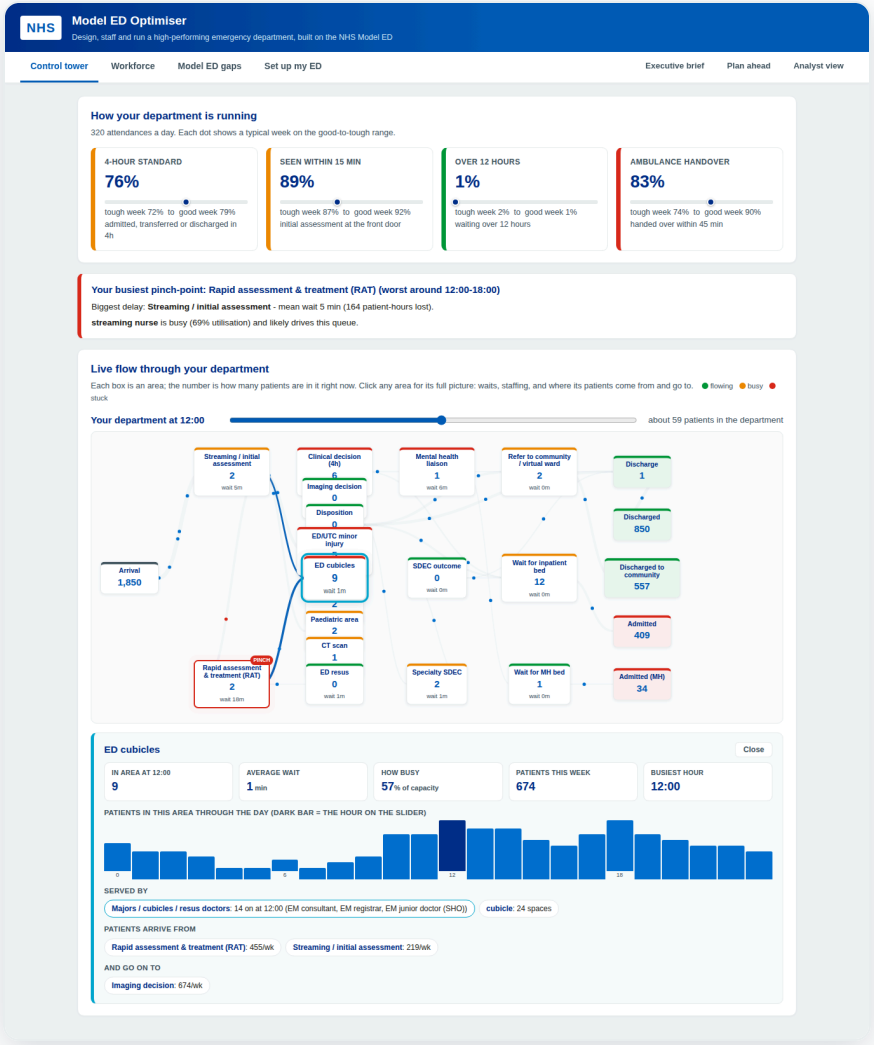
Today these are answered with spreadsheets, instinct and last year's pain. Each answer is a one-off, unauditable, and untested against reality.

Boards and ICBs increasingly ask for modelled evidence behind winter plans and capital bids. The trusts that bring it win the argument.

THE PRODUCT

Your department, as a live control tower

- Every area of **your ED** - your sizes, your opening hours, your rota
- Patients flow live; the **pinch-point pulses red** with when it bites
- Click any area: waits, staffing on shift, where patients come from and go to
- Headline standards as a **typical week with the good-week / tough-week spread**, never a false point estimate



WHY IT CAN BE TRUSTED

Three properties no spreadsheet has

Exit-block is emergent, not assumed

The model contains the hospital around the ED. When wards fill, admissions back up and corridor care appears - because the bed stock filled, not because a consultant typed a fudge factor. Turn the feedback off and the naive model returns.

It proves its own faithfulness

Every simulated patient journey is checked against the intended pathway (process-map conformance). A clean run passes at 100%; corrupted runs are caught. Clinical teams see the validation, not a promise.

Honest uncertainty, honest money

Every figure is a Monte-Carlo range, every pound comes from a trust-editable cost book with the assumptions printed on the page. Synthetic placeholders are flagged until your data replaces them.

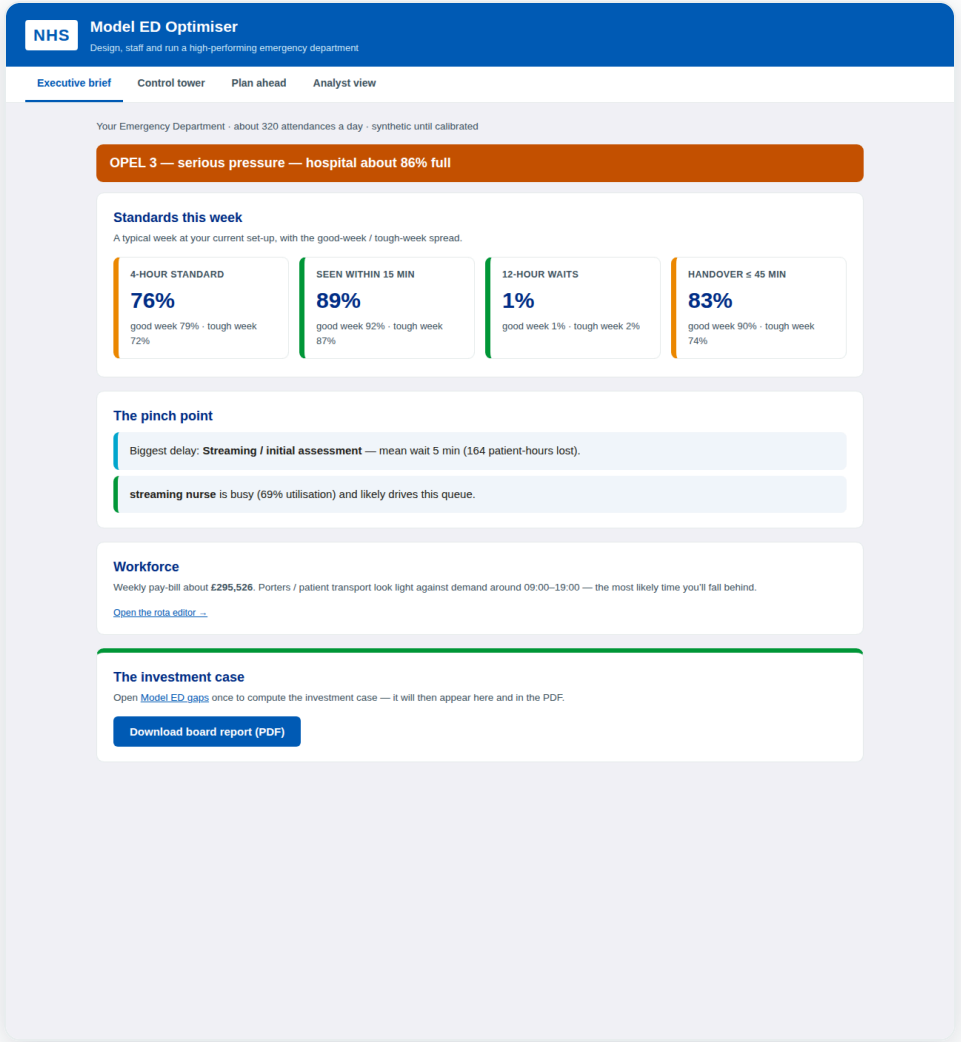
Real ED physics

Ambulance handover bays that block when the front door crowds. Patients who walk out when the wait beats their patience. Cubicles that close when the nursing rota thins. Porters who move admissions. Sickest first, always.

FOR THE EXECUTIVE

One screen, one honest position

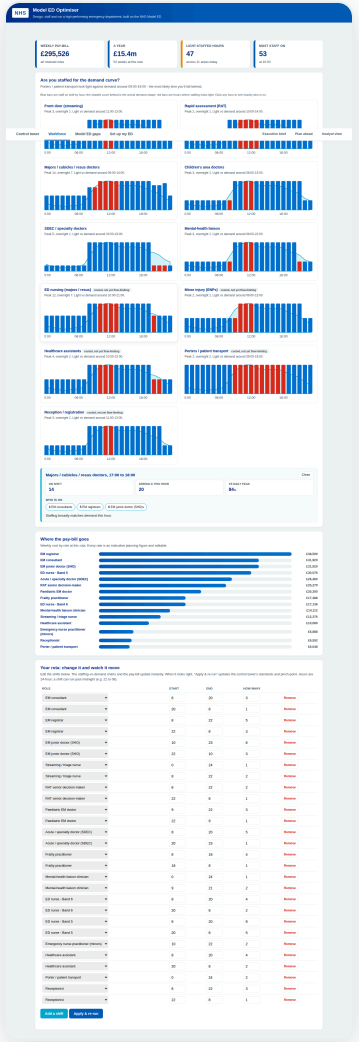
- **OPEL-style risk** from your simulated position and hospital fullness
- The four headline standards plus walk-outs, each with its spread
- **Where you sit against peers** - percentile position among similar-sized trusts
- The pinch point, the biggest investment case, and a **one-click board-ready PDF**



FOR THE WORKFORCE LEAD

Are you staffed for the demand curve?

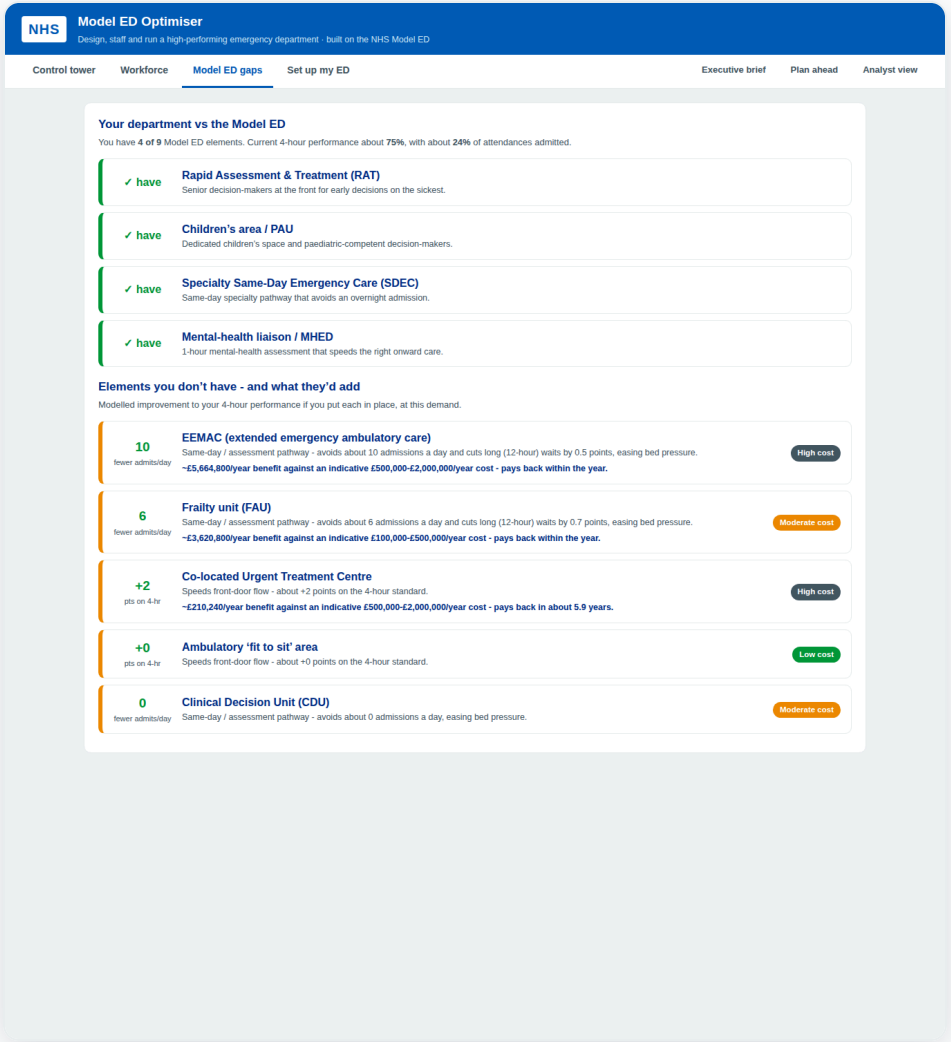
- Every role - consultants to porters - charted against the arrival curve, light hours in red
- Click an hour: **exactly who is on**, and the shift to add when it is light
- Edit the rota and watch the charts, the pay-bill and the pinch-point move **instantly**
- The full weekly pay-bill by role - the number a director of finance checks first



FOR THE BUSINESS CASE

Every gap priced, every claim backed by the model

- Which Model ED elements you have, which you lack, and what each would add **at your demand**
- Front-door elements judged on the 4-hour standard; same-day elements on **admissions avoided**
- Annual benefit, cost band, ROI and payback from **your** cost book, assumptions printed
- Flows into the board report PDF as a ready investment case



DECISION SUPPORT

Ask it the questions you actually have

"Would we cope with a bad flu season?"

Named stress tests - flu, norovirus ward closures, a strike day, heatwave - run against your ED exactly as configured. Then the inverse: the cheapest staffing additions that hold the 4-hour target, or an honest "staffing alone will not hold it".

"What should we change first?"

The recommendation engine searches your own weak points - rota gaps, pinch resources, opening hours - and only recommends changes whose improvement clears a statistical confidence bar. It would rather say nothing than recommend noise.

"What does this winter look like?"

Upload your daily history: the forecast learns your trend, weekday shape, seasonality and bank-holiday spikes - and shows its own backtested accuracy before you rely on it.

"Today is running hot - how bad by midnight?"

Type this morning's arrivals. It projects the day from your own curve, flags early escalation, and ranks the no-cost actions that help most tonight.

YOUR DATA, YOUR CONTROL

Built for NHS reality

- **Runs offline** - no cloud dependency, no CDN, deployable inside the trust network
- **No patient-identifiable data** - calibrates from aggregate ECDS-style extracts
- Every number traceable to the exact configuration that produced it
- Starts useful on day one with flagged synthetic defaults; gets sharper as your data replaces them

Calibrate once, everywhere

one upload updates the control tower, planner, exec brief and board report together

Self-validating

conformance checking runs with every simulation

Assumptions on every page

no invented benchmarks, no hidden fudge factors

ENGAGEMENT

A pilot you can see working in weeks

WEEKS 1-2

Discovery

Set up your ED in the product: areas, hours, rota, beds. It runs on day one.

WEEKS 3-4

Calibrate & validate

Your attendance extract replaces the placeholders; clinicians see the conformance validation.

WEEKS 5-8

Co-design scenarios

Winter plan, surge stress tests, workforce options - built with your ops and clinical teams.

WEEKS 9-12

The case & the habit

Board-ready business case from the model, and the control tower in weekly use.

Fixed-fee pilot → annual platform licence → optimisation cycles. The product does the analysis; your teams own the decisions.

THE ASK

Pick one department.
Give us twelve weeks.

We will hand your board a modelled winter plan, a costed investment case, and a control tower your site team checks every morning - all validated against your own data, with every assumption on the table.

Model ED Optimiser - see the queue before the queue sees you.